Erica F. Huang
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EDUCATION

Tufts University, Medford, MA **May 2027**
B.S.E. in Computer Science, Minor in Embedded Systems
3.88/4.0 GPA, Dean's List (5 semesters)

Relevant Courses: Data Structures, AI: Algorithms, Ethics, & Policy, Machine Structure & Assembly Language Programming, Embedded Systems, Human Computer Interaction, Digital Circuits, Technical Communication, Security

EXPERIENCE

MuLIP Lab (Tufts University)*, Research Assistant,* Medford, MA **Jan 2026 – Present**
- Contributing to developing task, movement, and grip policies to train robots to assist humans in open-world environments

Outreach Learning Fellows (Tufts University), *Fellow,* Medford, MA **Jan 2026 – Present**
- Providing engineering experiences to 5th grade students from diverse backgrounds who might otherwise lack access to hands-on engineering

Human Robot Interaction Lab (Tufts University), *Research Assistant,* Medford, MA **May 2025 – Present**
- Running and measuring experiments to understand how teams work with the goal of creating an LLM to assist teamwork; expected to be featured in the resulting research paper

Cytel, *Software Engineering Intern*, Wellesley, MA **Summer 2024, Summer 2025**
- Optimized code using parallel processing and communicating between R and C++ for faster runtime to speed up running clinical trial simulations
- Generated reports in R using simulation output to provide clients with a formatted summary of their simulated trials
- Tested different forms of optimization (using a dynamically linked library to communicate between C++ and R, running C++ in parallel, and running R in parallel) to understand which method results in the fastest runtime
- Presented well-received analysis and recommendations to a team of senior developers for further development of my work

FIRST LEGO League (FLL) Team, *Founder and Lead Student Mentor*, Wellesley, MA **May 2022 – May 2023**
- Created and led a partnership with Wellesley Middle School science and technology teachers to launch two FIRST LEGO League Challenge teams for 6th through 8th-grade students, providing them free access to hands-on engineering
- Placed 9th and 19th out of 36 teams at the 2022 regional competition
- Won the team award for displaying FIRST core values

FIRST Robotics Competition (FRC), *Founding Member and Coding Lead,* Wellesley, MA **Aug 2020 – May 2022**
- Led a nine-member coding team, contributing to an award-winning robot
- Expanded and gender-diversified coding team, contributing to a second award-winning robot
- Established and served in the team's first Director of Community Education and Outreach role
- Created and implemented summer robotics workshops for 3rd and 4th graders and collaborated with local Girl Scout leaders to develop and run badge-earning events

PROJECTS

Guitar Hero: Used System Verilog to program a basic version of Guitar Hero to run on an FPGA. Designed and implemented game logic, game registers, and graphics (displayed on a VGA monitor and adapter) with a group.
UM: Used C to implement a UM emulator and its memory architecture with a partner. Runs using 14 different instructions, including those that read to and write from memory, and those that load programs to run.
Reverse Polish Notation (RPN) Calculator: Used a course-given assembly language to implement an RPN calculator with a partner. The program includes memory space allocation, a jump table, a call stack, a value stack, and registers.

SKILLS

Programming Languages: C++, C, Python, Java, R, LaTeX, HTML, CSS, System Verilog, Impcore, Scheme, ARM
Software: Linux, Git, multithreading, VS Code, Wireshark, debugging tools, Microsoft Office
Hardware: FPGA, microcontroller, breadboard, voltmeter, oscilloscope, VGA monitor

ACTIVITIES

Hobbies: Snowboarding, weightlifting, basketball, triathlons, rock climbing, LEGOs, backpacking, puzzles, saxophone
Tufts Clubs: Basketball (social chair), Wilderness Orientation (leader), Mountain Club, Robotics, Mechanical Keyboards